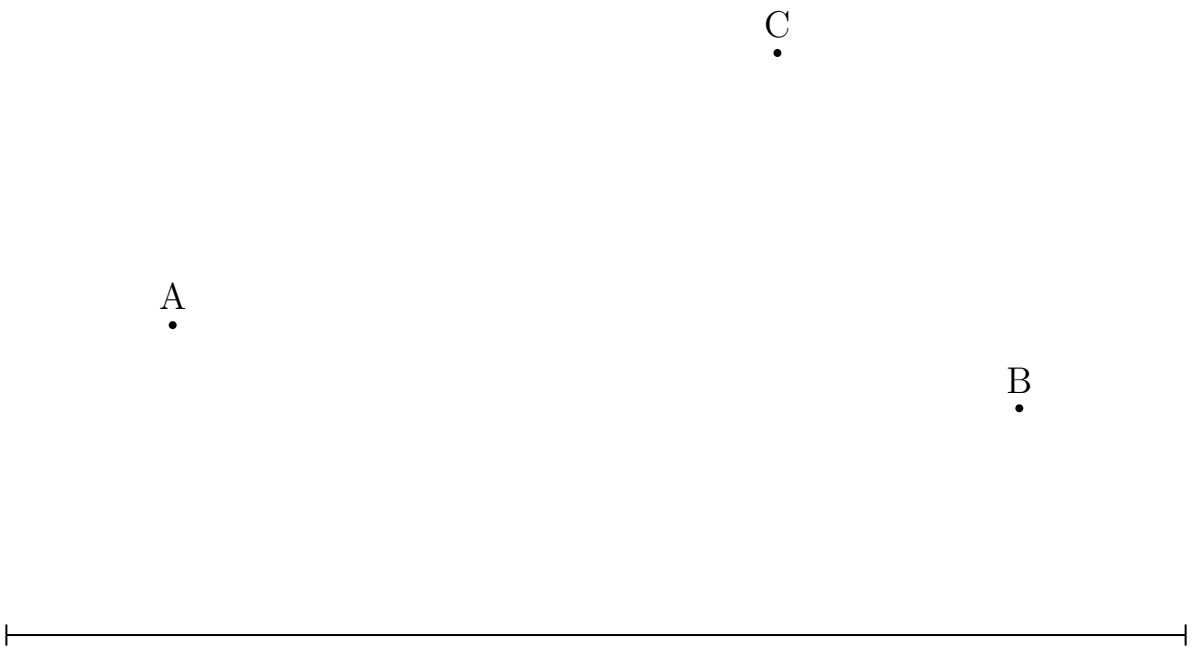
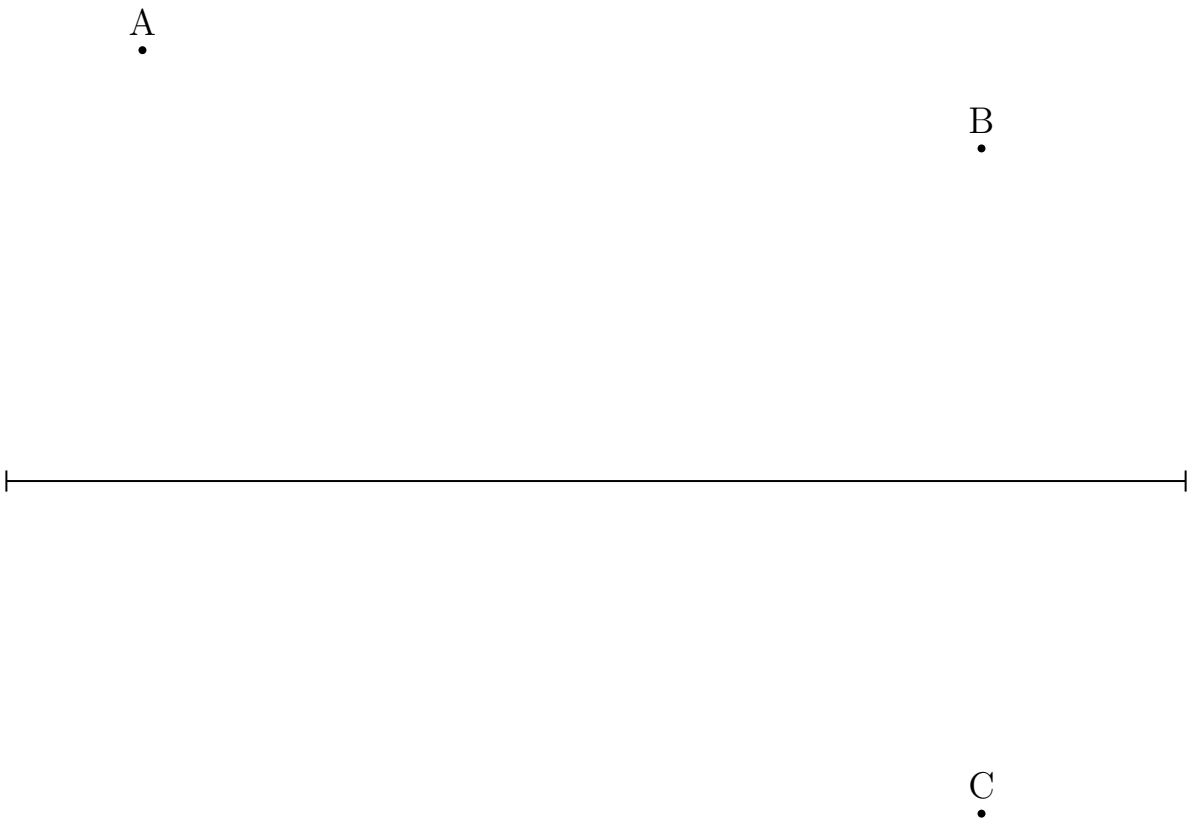
A route goes straight from its start dot to one point on the line, then straight to its finish dot. Touch between the end marks. Do not travel along the line. Use the dot centers.

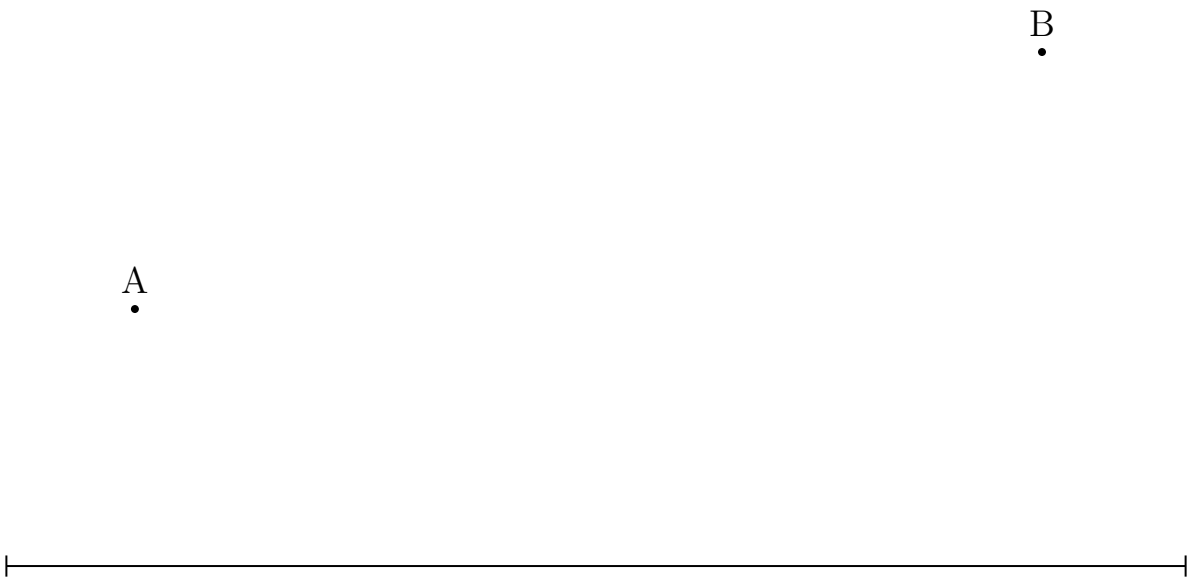
Problem 1: Find the shortest routes from A to B and from A to C. Which route needs less string?



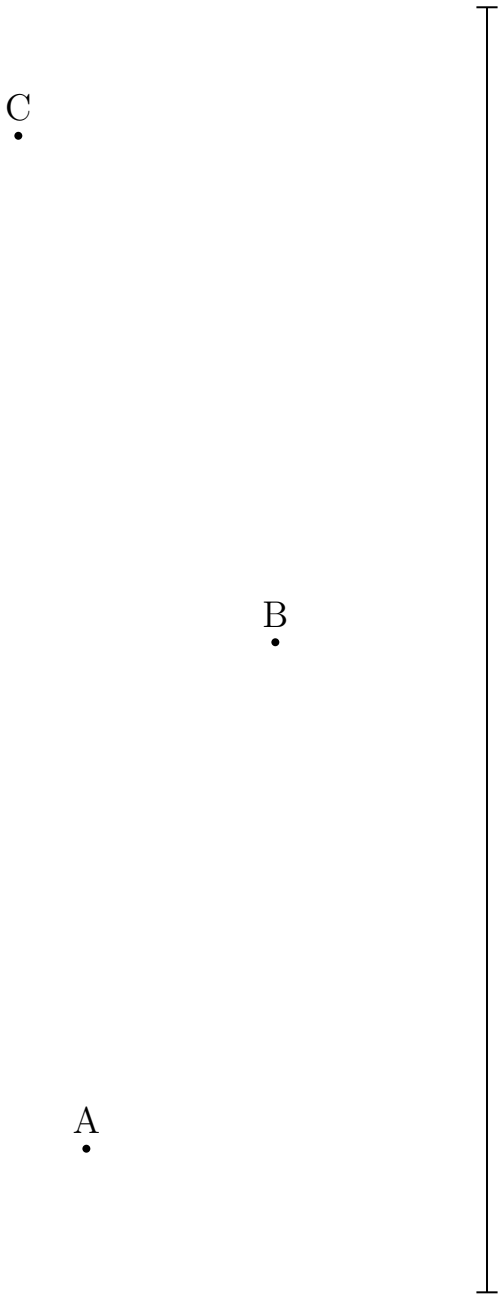
Problem 2: Make routes from A to B and from A to C that touch the line at the same point. Could moving that point make one route longer than the other? Explain.



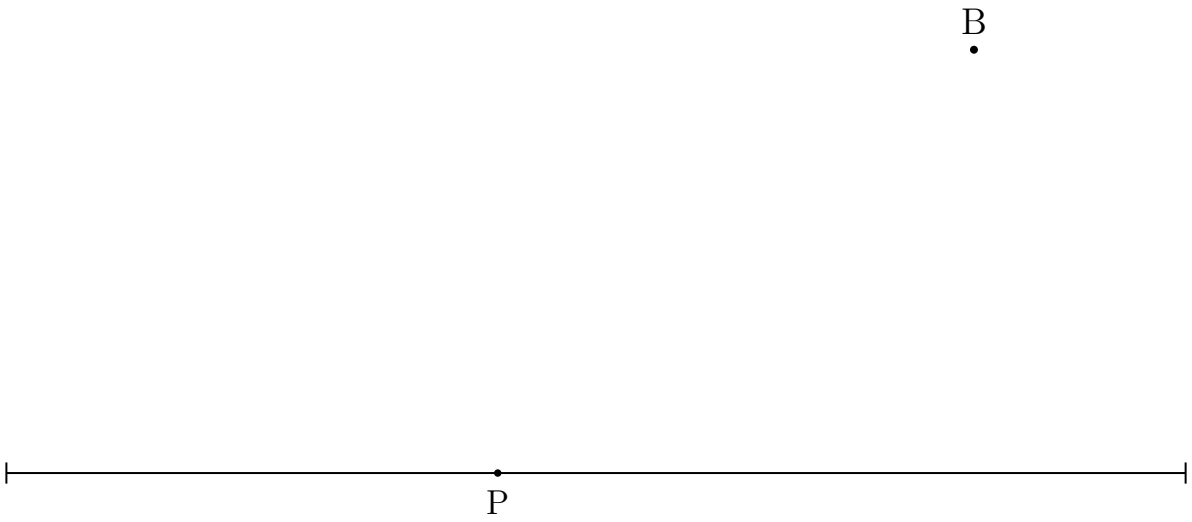
Problem 3: Find the shortest route from A to B. Explain why no other contact point gives a shorter route.



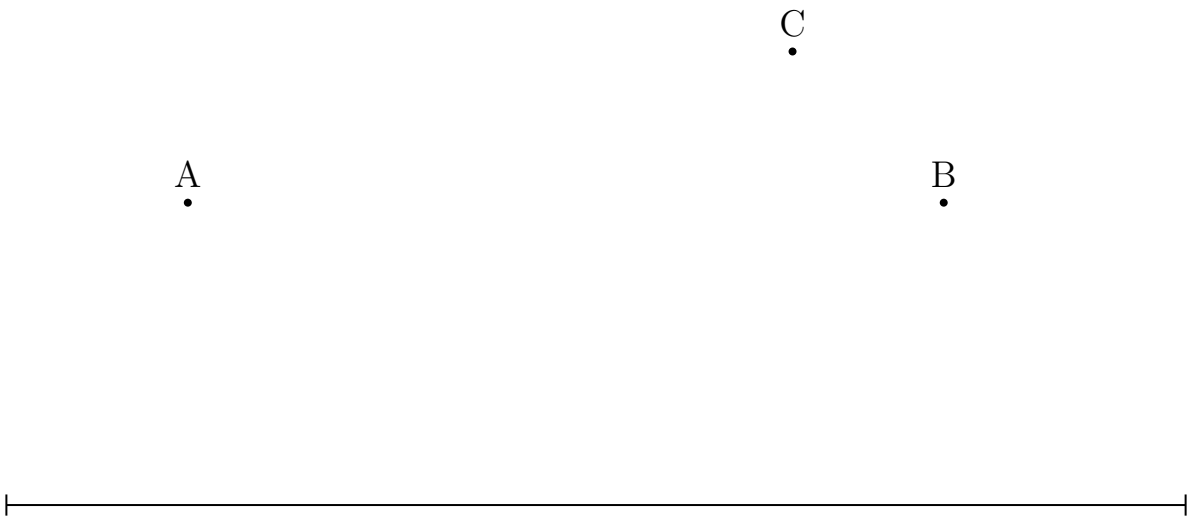
Problem 4: Find the shortest route for each pair: A to B, A to C, and B to C.



Problem 5: Choose two different start dots above the line so that their shortest routes to B both touch P. Draw both routes.



Problem 6: Find the shortest routes from A to B and from A to C. Which needs less string, or do they need the same amount?



Problem 7: Find the shortest route from A to B. Now allow touching only between C and D, only between E and F, or only between C and F. In which cases does the same route still give a shortest allowed route? Explain.

